

PROCESSOR

Processor

- **Central Processing Unit (CPU)**, or sometimes just called **processor**
- **The CPU is the brains of the computer.**
- is similar to a calculator, only much more powerful and the main function of the CPU is to perform arithmetic and logical operations on data taken from memory or on information entered through some device, such as a keyboard, scanner, or joystick.



INTEL

Intel Corporation

- is the world's largest semiconductor company and the inventor of the x86 series of microprocessors, the processors found in most personal computers.
- Founded on July 18th, 1968 as **Integrated Electronics Corporation**.
- Intel also makes motherboard chipsets, network cards and ICs, flash memory, graphic chips, embedded processors, and other devices related to communications and computing.



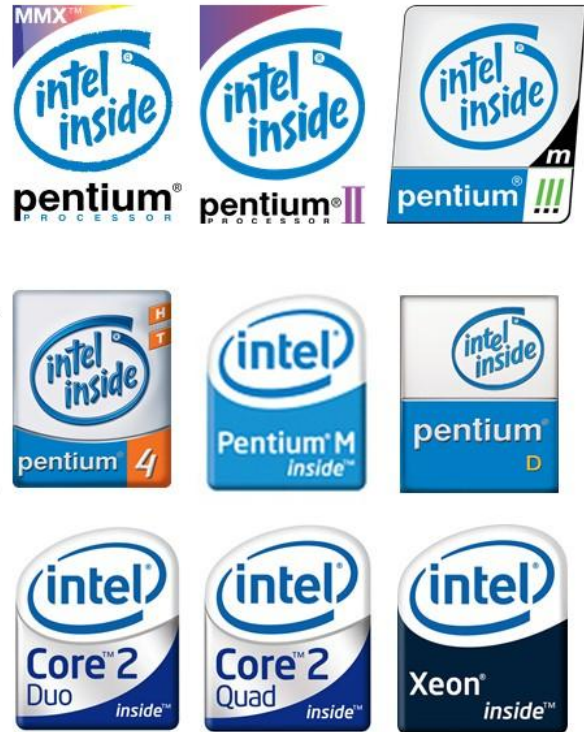
Intel headquarters
in Santa Clara,
California



Intel microprocessor
facility in Costa Rica

Pentium

- Introduced on March 22, 1993
- Is a brand refers to Intel's single-core x86 microprocessor
- term **x86** refers to the instruction set
- Intel selected the *Pentium* name after courts had disallowed trademarking of names containing numbers - like "[286](#)", "[i386](#)", "[i486](#)"
- Starting in 1995, Intel (inconsistently) used the "Pentium" registered trademark



Celeron

- Introduced in April 1998
- brand refers to a range of Intel's x86 CPUs for budget/value personal computers
- Celeron processors have certain high-end processor features disabled (like dual processing or multiprocessing).
- The Celerons are also slower than similar-clocked Pentiums due to smaller L2 cache size and sometimes slower bus speed.



Advanced Micro Devices, Inc.

- is an American multinational semiconductor company based in Sunnyvale, California
- It develops microprocessors, motherboard chipsets, embedded processors and graphics processors for servers, workstations and personal computers, and processor technologies for handheld devices, digital television, and game consoles.



AMD headquarters in Sunnyvale, California

AMD Markham, which used to be the ATI headquarter.



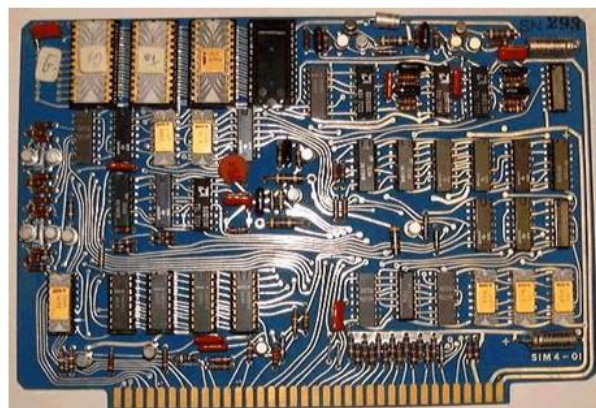
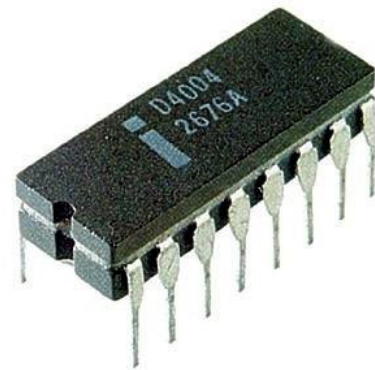
AMD



Intel Processor

Intel 4004

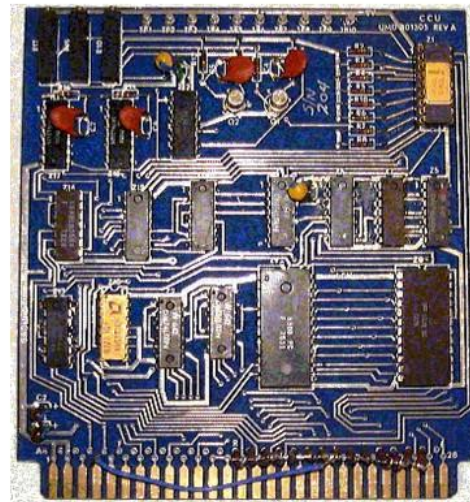
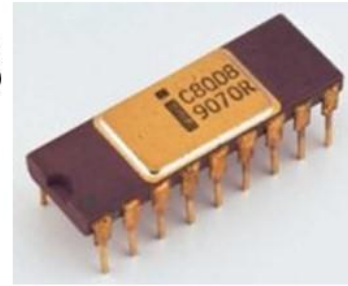
- is the first computer processor designed and manufactured by chip maker Intel
- is a 4-bit central processing unit (CPU) released in 1971
- **Max CPU clock:** 740 kHz
- **Package:** 16 pin



4004 Board

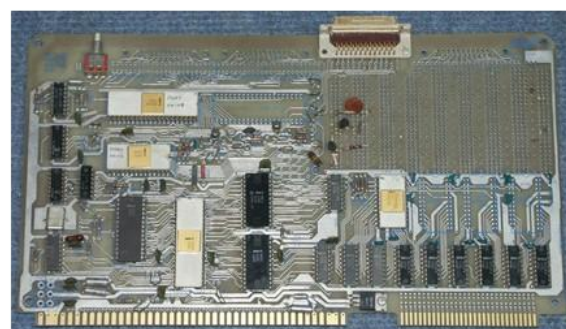
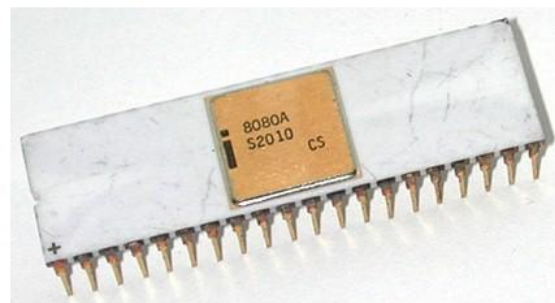
Intel 8008

- introduced in April 1972.
- was twice as powerful as the 4004
- **Max CPU clock:** 0.5 Mhz to 0.8 Mhz
- **Package:** 18 pin DIP



Intel 8080

- 8-bit CPU
- was released in April 1974
- **Max CPU clock:** 2 MHz
- **Package:** 40 pin DIP
- was used in many early microcomputers, such as the MITS Altair 8800



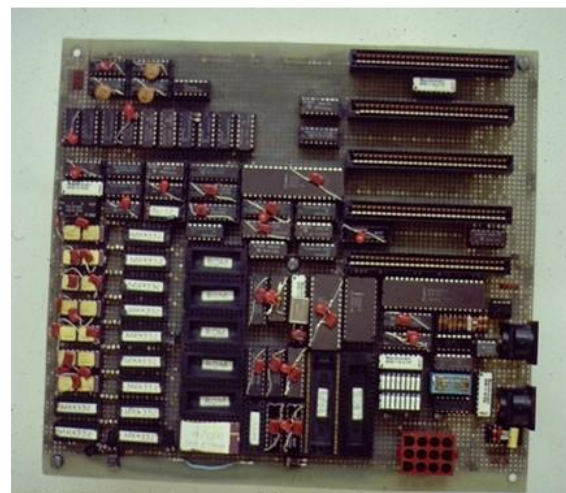
Intel 8086

- released in 1978
- 40 pin [DIP](#)
- Internal clock speed is from 4.77 to 8 Mhz



Intel 8088

- was introduced on July 1, 1979
- 40 pin DIP
- Internal clock speed is from 4.77 to 8 Mhz



Intel 80286

- introduced on February 1, 1982
- 68-pins
- Internal clock speed is from 6 to 20 Mhz



Intel 80386 DX

- was released by [AMD](#) in [1991](#)
- A [100%-compatible clone](#) of the [Intel 80386](#) design
- 132-pins
- Internal clock speed is from 12 to 40 Mhz



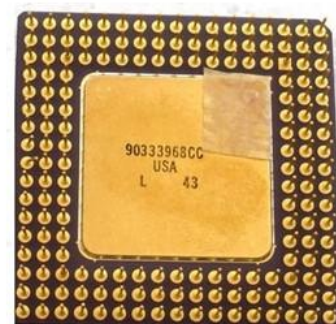
Intel 80386 SX

- otherwise known as the **Intel386**, **i386** or just **386**
- Introduced in 1988
- 132-pin PGA
- Internal clock speed is from 16 to 33 Mhz



Intel 80486 DX

- was introduced in 1989 containing 1.2 million transistors
- Internal clock speed is from 25 to 50 Mhz
- 168 pins PGA



Intel 80486 SX

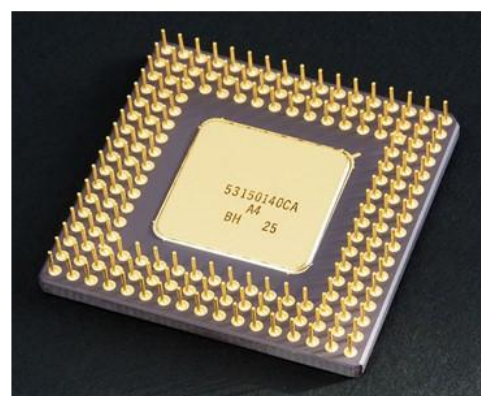
- was introduced in 1989
- was a modified [Intel 486DX microprocessor](#)
- Internal clock speed is from 25 to 50 Mhz
- 168 pins PGA



Intel 80486 DX2



- was introduced in 1992
- had 168 pins
- Internal clock speed is from 50 to 80 Mhz



Intel 80486 DX4

- Was introduced in March 7, 1994
- 168 -pin PGA (pin grid array)
- Internal clock speed is from 75 to 120 Mhz

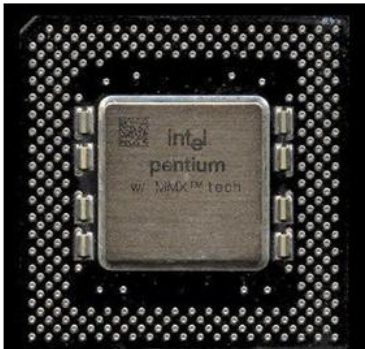


Pentium

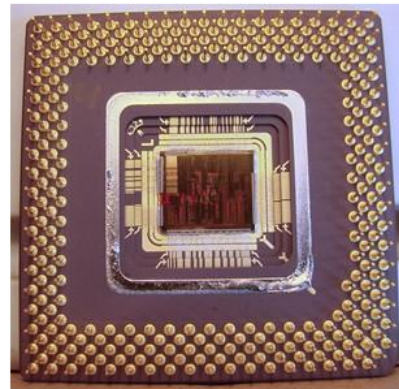
- Introduced March 22, 1993
- Socket 4, Socket 5, Socket 7
- 273 pin PGA processor
- Internal clock speed is from 60 to 200 Mhz



Pentium MMX



- Introduced January 8, 1997
- Socket 7
- 296/321 pin PGA (pin grid array)
- Internal clock speed is from 166 to 233 Mhz



Pentium Pro



- Introduced November 1, 1995
- Socket 8 processor



- 387 pins
- Internal clock speed is from 150 to 200 Mhz



Pentium II

- introduced May 7, 1997
- Slot 1 / 242 Pins
- **Max CPU clock** 233 MHz to 450 MHz



Pentium III

- Introduced February 26, 1999
- 242-pin Slot 1
- Slot 1

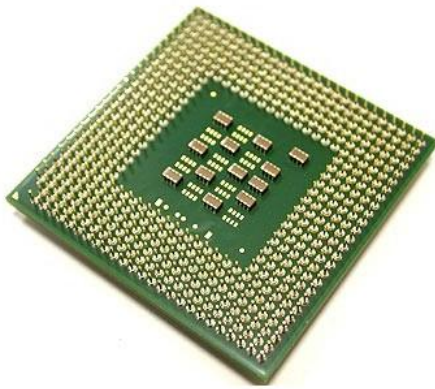


- 370-pin PGA
- Socket 370
- **Max CPU clock** 450 MHz to 1.4 GHz



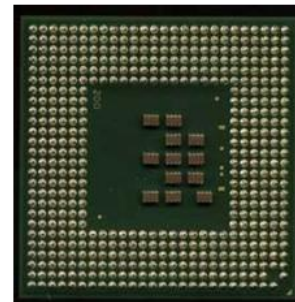
Pentium 4

- Introduced November 20, 2001
- PGA423, PGA478
- Had 423 and 478 pins
- **Max CPU clock 1.3 GHz to 3.8 GHz**

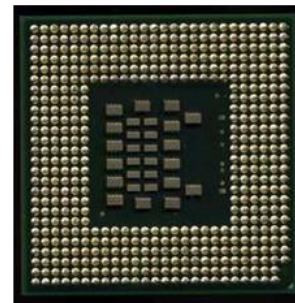


Pentium M

- introduced in March 2003
- The first Pentium M branded CPU, codenamed the **Banias**, was followed by the second one - the **Dothan**. **After the Pentium M branded processors, Intel released the Core branded dual-core mobile Yonah**
- **Socket:** Socket 479
- were intended for use in laptop personal computers
- Despite the 479 in the name the Pentium M Processors for this socket use only 478 Pins
- **Max CPU clock:** 900 MHz to 2.26 GHz



Dothan (Pentium M SL7SM)



Yonah (Core Duo)

Intel Pentium D

- refers to two series of dual-core
- Introduced May 25, 2005
- LGA 775
- Internal clock speed is from 2.66 GHz to 3.73 GHz



Intel Core 2

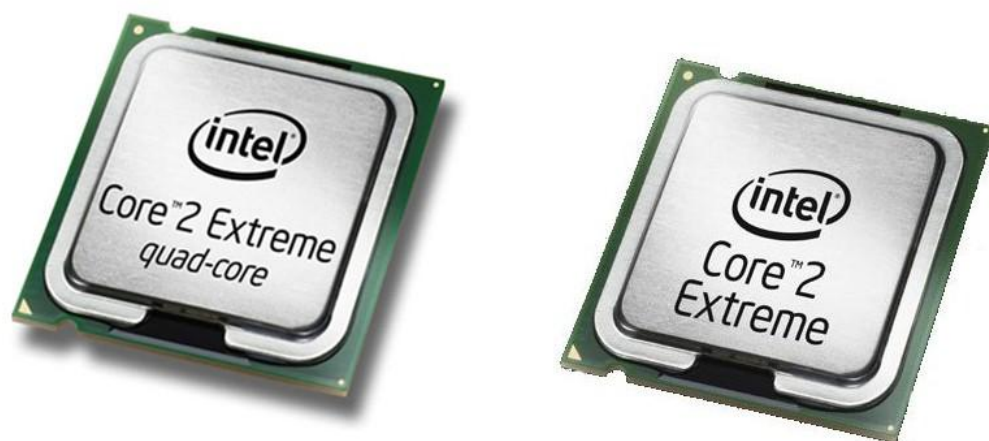
- introduced on [July 27, 2006](#)
- **Max CPU clock:** 1.06 [GHz](#) to 3.33 [GHz](#)
- Socket T, Socket M, Socket P, Micro-FCBGA



Intel Core 2 Duo



Intel Core 2 Extreme



AMD Processor

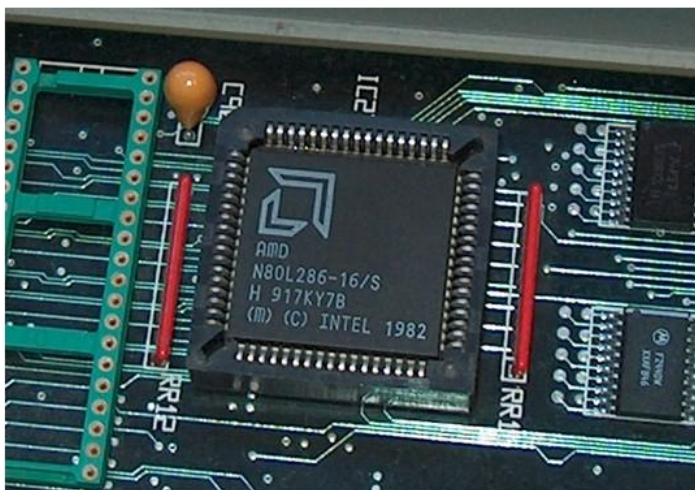
Am9080

- It was originally produced without license as a clone of the Intel 8080
- The first versions of the Am9080 available in April of 1974.
- This CPU operated at a speed of 2 MHz



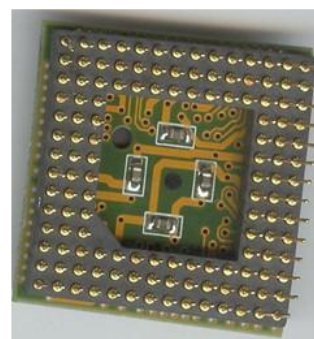
Am286

- Am286, a clone of the [Intel 80286](#) manufactured under license, was identical to the chip from Intel, but it had a big advantage: its higher clock speed.
- The chip was later sold by AMD as an embedded processor
- Clock frequency 8-20 MHz



Am386 DX

- The **Am386** CPU was released by AMD in 1991.
- A 100%-compatible clone of the Intel 80386 design
- While Intel's 386 design peaked at 33 MHz, AMD released a 40 MHz version of both its 386DX and 386SX



Am486

- is a 80486-class family of computer processors that was produced by AMD in the 1990s.
- This CPU operated at a speed of 25 MHz to 120 MHz



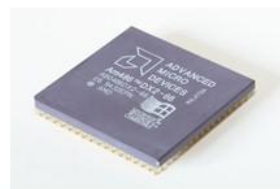
AMD Am5x86 75MHz



Am486DE2-66 with power management for embedded systems.



AMD Am486DX 40 MHz



AMD Am486DX2 66MHz



AMD Am486SX2 66MHz

Am5x86

- Introduced in November 1995
- is an x86-compatible CPU introduced in 1995 by AMD for use in 486-class computer systems.
- It was one of the fastest, and most universally-compatible upgrade paths for users of 486 systems.
- This CPU operated at a speed of 75 MHz to 200 MHz



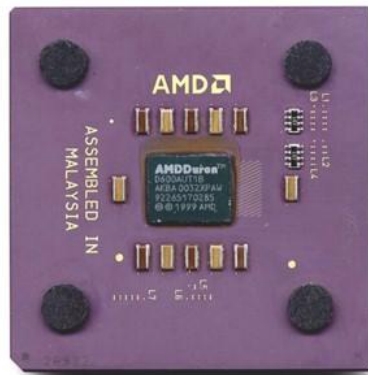
Am5x86- 75MHz for Socket 3



Am5x86- 75Mhz surface mount

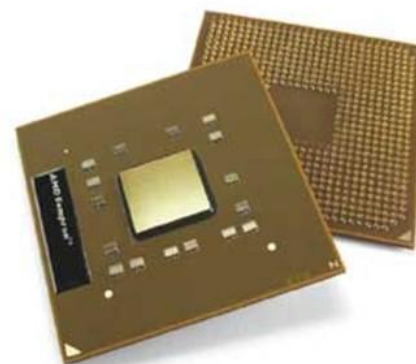
Duron

- It was released on June 19, 2000 as a low-cost alternative to AMD's own Athlon processor and the Pentium III and Celeron processor lines from rival Intel.
- The Duron was discontinued in 2004 and succeeded by the Sempron.
- **Max CPU clock** 600 MHz to 1.8 GHz
- Socket A



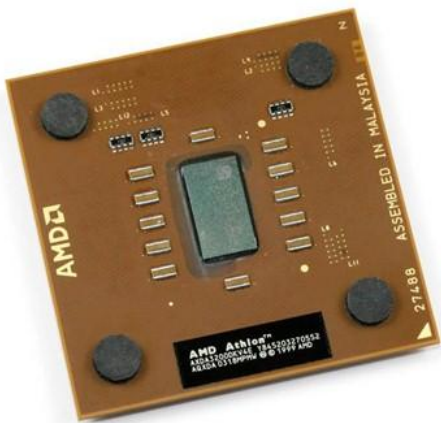
Sempron

- The **Sempron** is AMD's newest low-end CPU, replacing the Duron processor.
- Socket A, 754, 939, AM2, S1
- Cpu speed is from 1.5 to 2.2 MHz



ATHLON

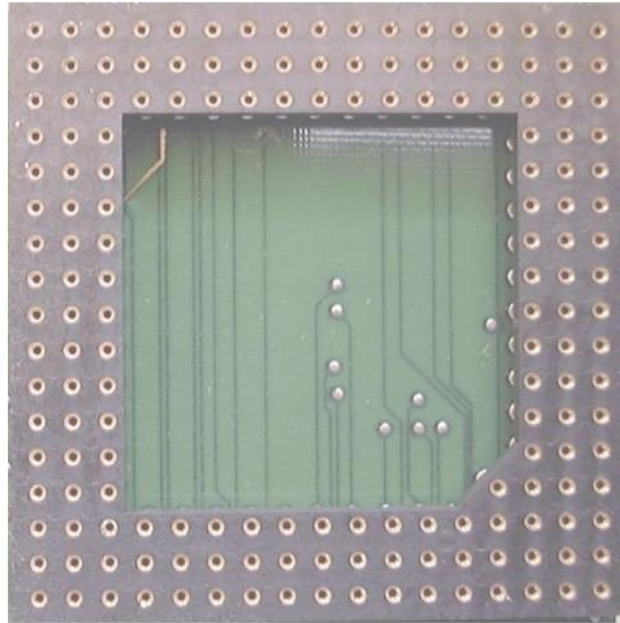
- launched on June 23, 1999.
- Athlon Classic is a cartridge-based processor. The design, called Slot A, was quite similar to Intel's Slot 1 cartridge used for Pentium II and Pentium III
- **Max CPU clock** 500 MHz to 2.33 GHz
- Slot A
Socket A



Intel CPU Interface

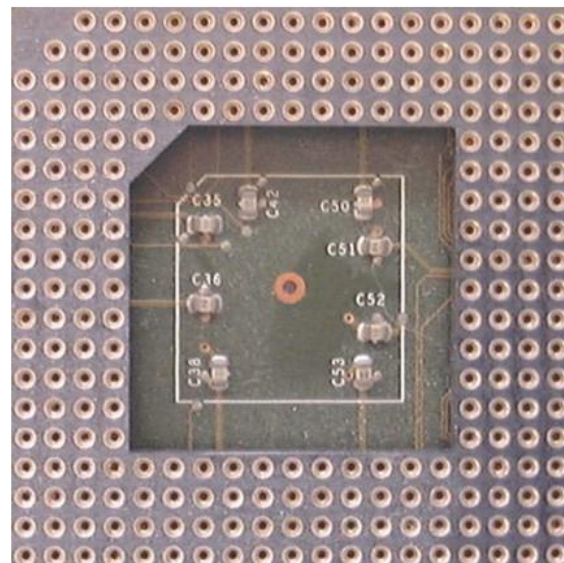
Socket 1

- was a 169 pin
- Found on 486 motherboards
- suitable for the 16 - 33 MHz 486 SX, 486 DX, 486 DX21 and DX4 processors.



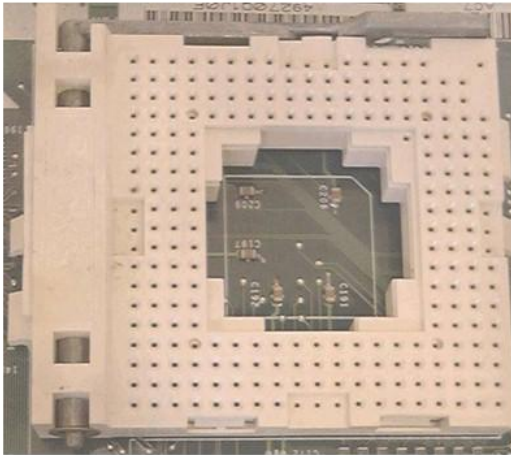
Socket 2

- **Number of Pins:** 238
- A minor upgrade from socket 1 that supported all the same chips.
- **Supported Processors:** 486SX, 486DX, 486DX2, 486DX4 OverDrive, Pentium OverDrive



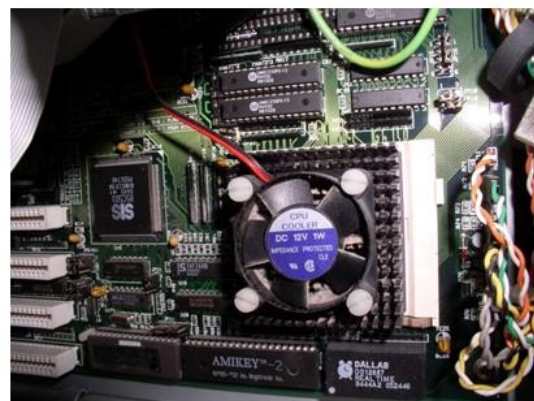
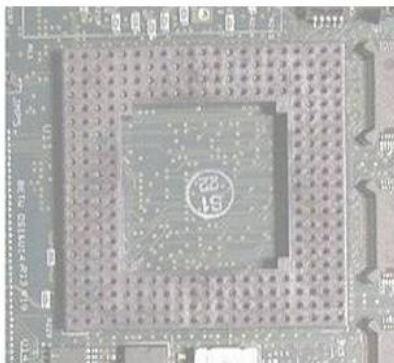
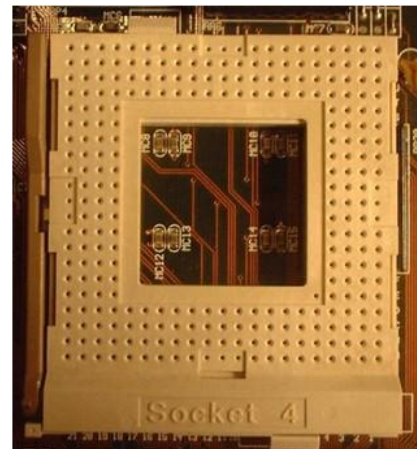
Socket 3

- was a 237 pin
- Support all of socket 2 chips
- **Supported Processors:**
486SX, 486DX, 486DX2,
486DX4, Pentium OverDrive



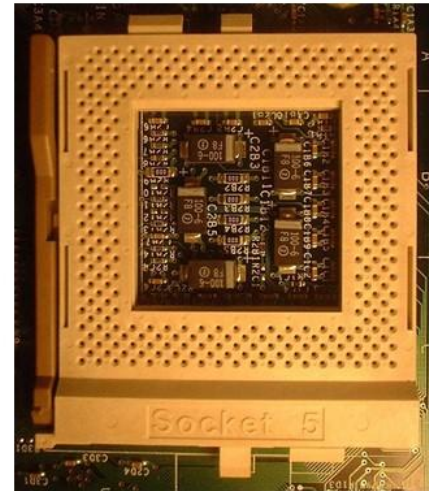
Socket 4

- presented in 1993
- first socket designed for native support of the early Pentium processors, running at 60 or 66 MHz
- Number of Pins: 273
- **Supported Processors:** Pentium 60-66, Pentium OverDrive



Socket 5

- was created for the second generation of Intel Pentium processors operating at speeds from 75 to 133 MHz as well as certain Pentium OverDrive processors
- Pentium MMX processors are not compatible with Socket 5
- Consisting of 320 pins

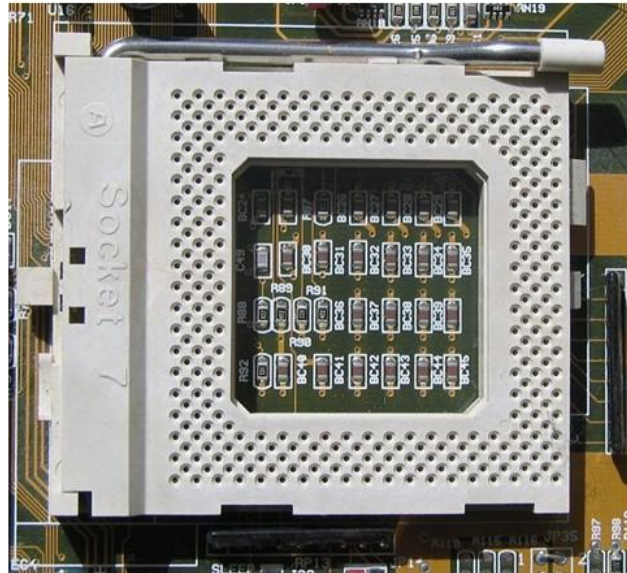


Socket 6

- was a 486-generation CPU socket, a slightly modified version of the more common Socket 3
- **Number of Pins:** 235 pin
- **Supported Processors:** 486DX4, Pentium OverDrive

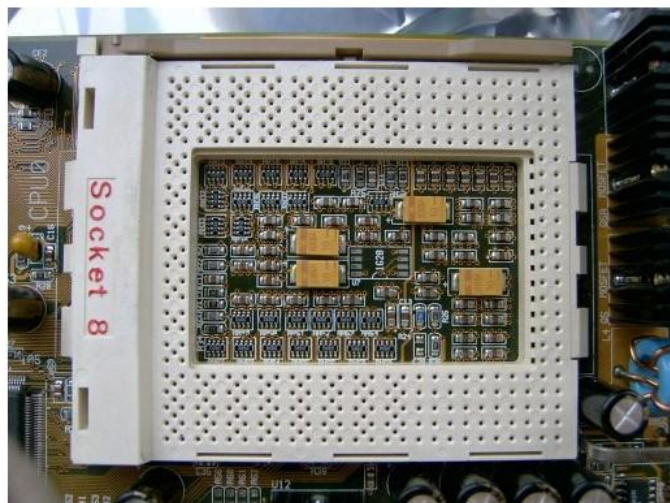
Socket 7

- **Contacts** :321 pins
- **FSB** : 66 - 83 MHz System Clock
- **Processors** :75-233 MHz Intel Pentium, AMD K5 through K6, Cyrix 6x86 (and MX) P120 - P233



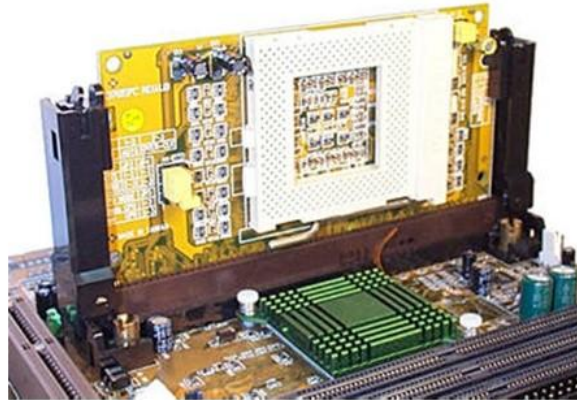
Socket 8

- was used exclusively with the [Intel Pentium Pro](#) and [Pentium II Overdrive computer processors](#).
- Socket 8 has a unique rectangular shape with 387 pins



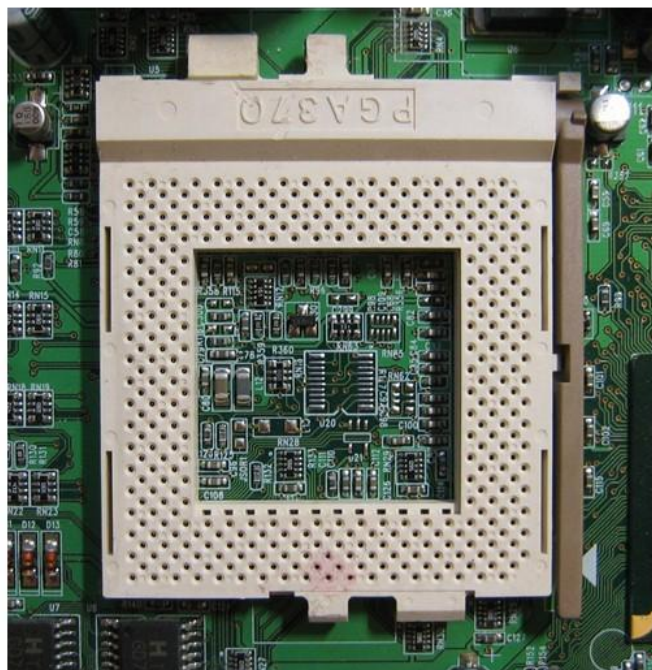
Slot 1

- refers to the physical and electrical specification for the connector used by some of Intel's microprocessors, including the Celeron, Pentium II and the Pentium III. Both single and dual processor configurations were implemented.
- much like a PCI slot, but with a 242-lead edge-connector.



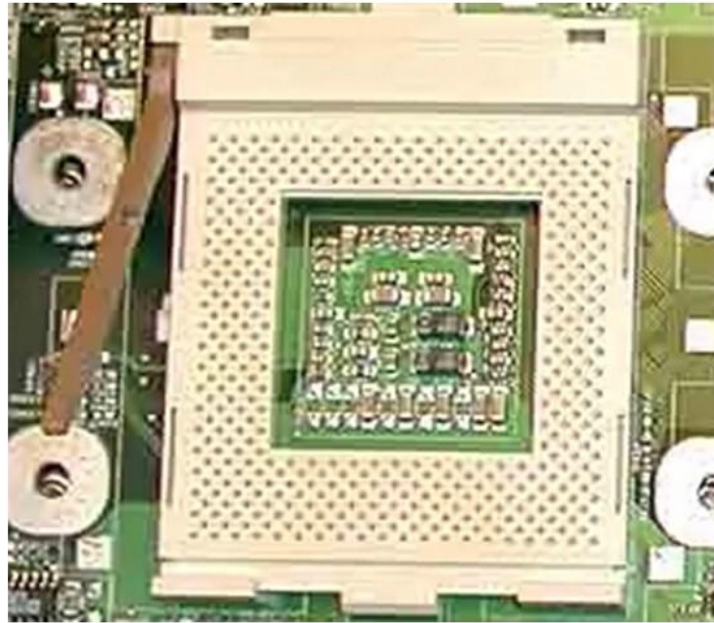
Socket 370

- also known as the PGA370 socket
- is a common format of [CPU socket](#) first used by [Intel](#) for [Pentium III](#) and [Celeron](#) processors to replace the older [Slot 1](#) CPU interface on [personal computers](#).
- The "370" refers to the number of holes in the socket for CPU pins.



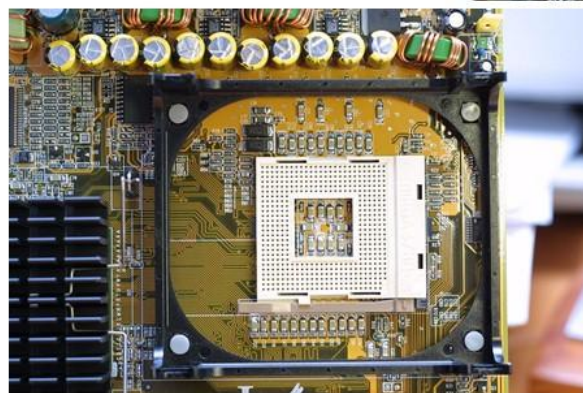
Socket 423

- used for the first [Pentium 4 processors](#)
- The socket was short-lived, as it became apparent that its electrical design proved inadequate for raising [clock speed](#) beyond 2.0 GHz.
- It has 321 pin



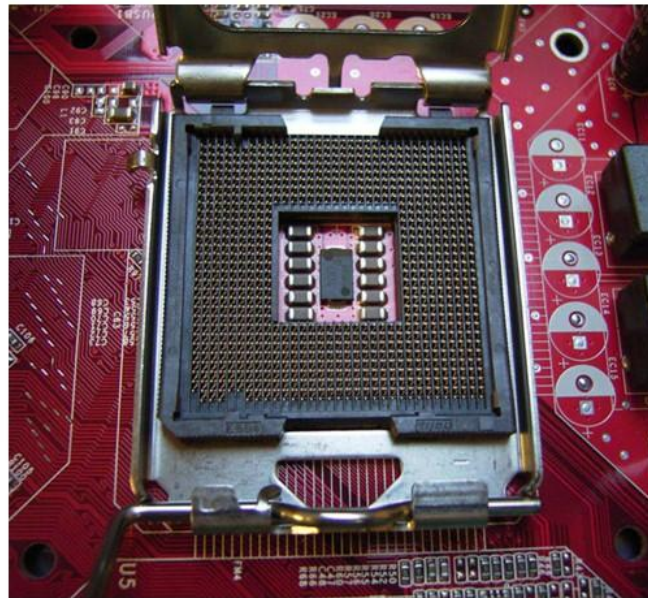
Socket 478

- is a type of CPU socket used for Intel's Pentium 4 and Celeron series CPUs.
- Motherboards that use this socket support DDR, RDRAM, and in some cases SDRAM.



Socket T Land Grid Array

- also known as LGA775
- has no socket holes, instead it has 775 protruding pins which touch contact points on the underside of the processor (CPU)
- The 'T' in Socket T was derived from the now cancelled Tejas core
- Tejas was a code name for Intel's microprocessor which was to be a successor to the latest Pentium 4
- Support Intel Pentium 4, Pentium D, Celeron D, Pentium Extreme Edition, Core 2 Duo, Core 2 Extreme, Xeon 3000 series, Core 2 Quad



AMD CPU Interface

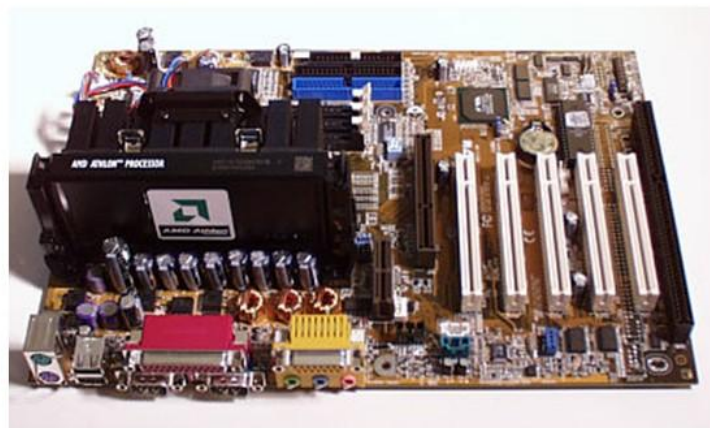
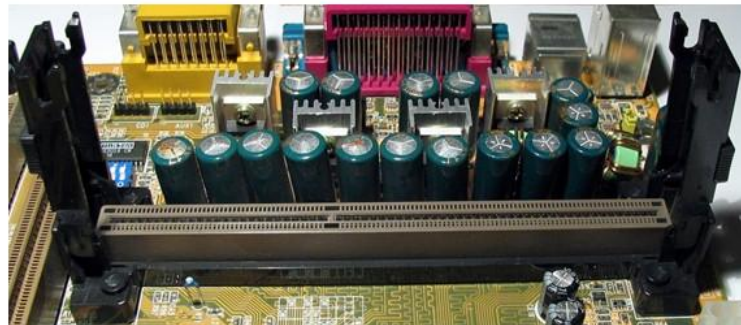
Socket 7

- also referred to as **Super Socket 7**
- Had 321 pins / contacts
- used by AMD K6-2
AMD K6-III
AMD K6-2+
AMD K6-III+
Processors



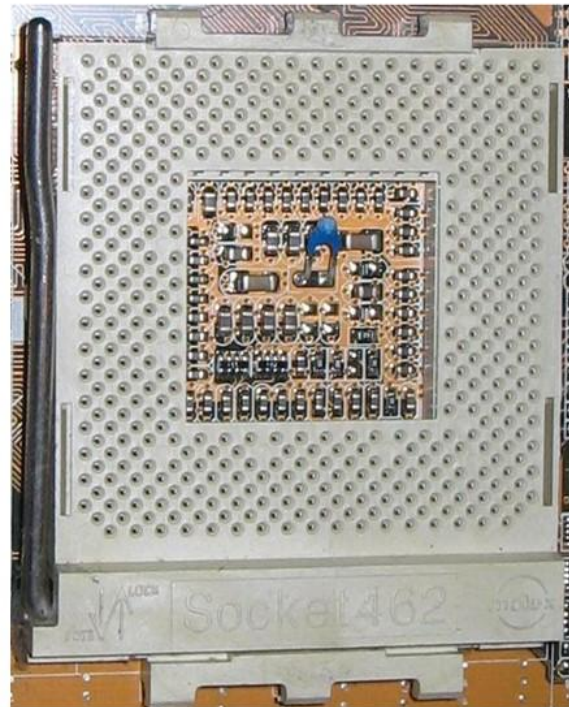
- refers to the physical and electrical specification for the edge-connector used by early versions of [AMD's Athlon](#) processor
- is a 242-contact slot that accepts a microprocessor packaged as a Single Edge Contact (SEC) cartridge.
- [AMD's Athlon](#) processor 650 –1000 MHz

Slot A



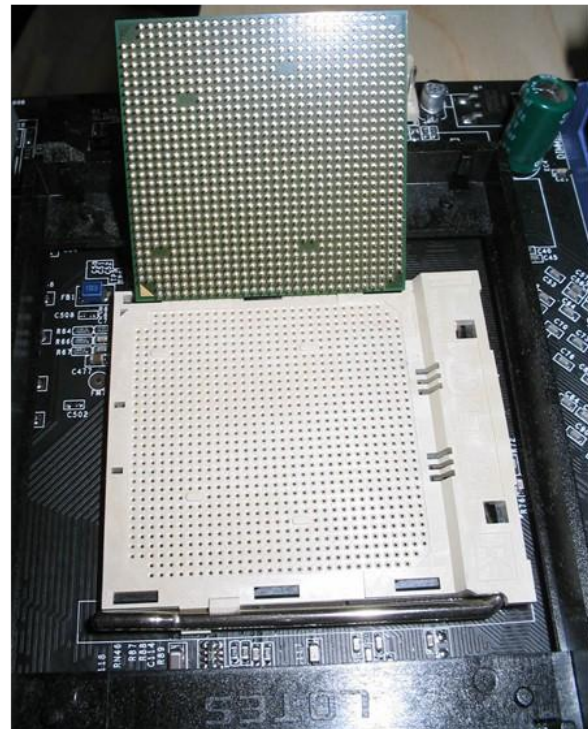
Socket A

- also known as **Socket 462**
- is the CPU socket used for AMD processors ranging from the Athlon Thunderbird to the Athlon XP/MP 3200+, and AMD budget processors including the Duron and Sempron.
- is a zero insertion force pin grid array type with 453 pins
- **Processors**
 - AMD Athlon (650 MHz – 1400 MHz)
 - AMD Athlon XP (1500+ – 3200+)
 - AMD Duron (600 MHz – 1800 MHz)
 - AMD Sempron (2000+ – 3300+)
 - AMD Athlon MP (1000 MHz – 3000+)
 - AMD Geode NX (667 MHz – 2200 MHz)



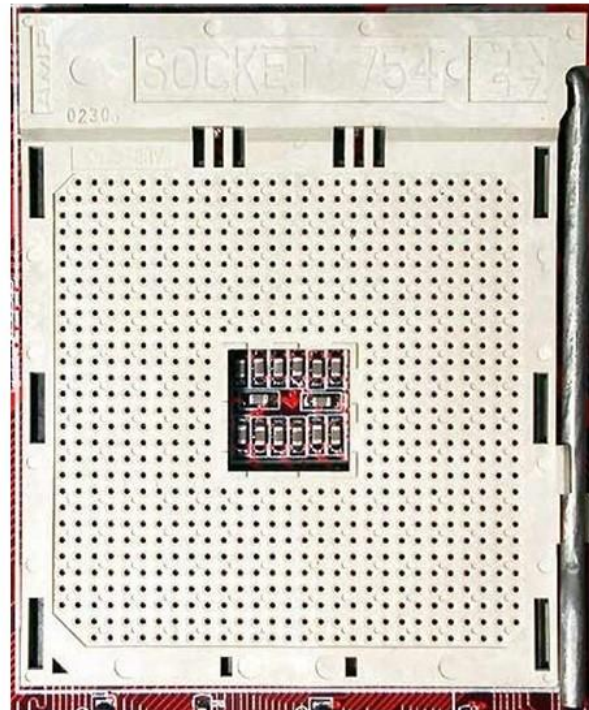
Socket 939

- is a CPU socket released by AMD in June 2004 to supersede the previous Socket 754 for Athlon 64 processors.
- It has 939 pins
- **Processors**
 - AMD Athlon 64 (3000+ - 4000+)
 - AMD Athlon 64 FX (51 - 60)
 - AMD Athlon 64 X2 (3600+ - 4800+)



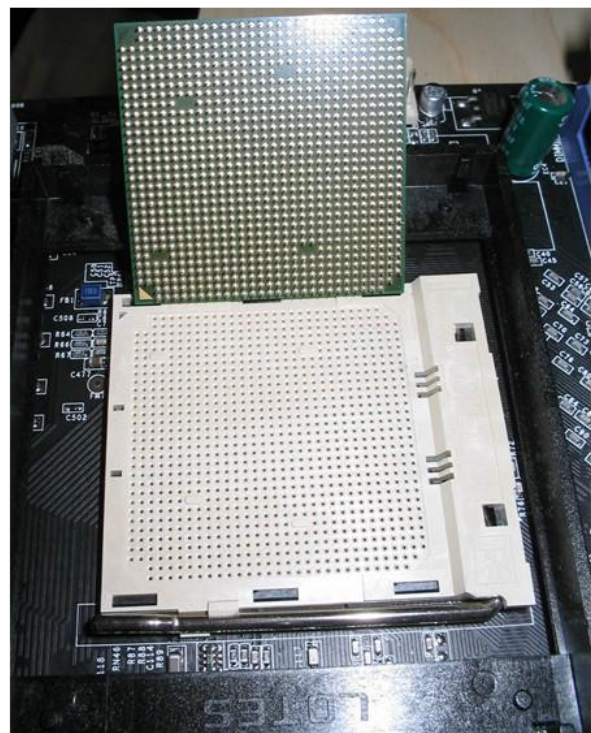
Socket 754

- Socket 754 was the first socket developed by AMD to support their new consumer version of the 64 bit microprocessor family known as AMD64.
- **Contacts 754**
- **Processors**
 - AMD Athlon 64 (2800+ - 3700+)
 - AMD Sempron (2500+ -)
 - AMD Turion 64



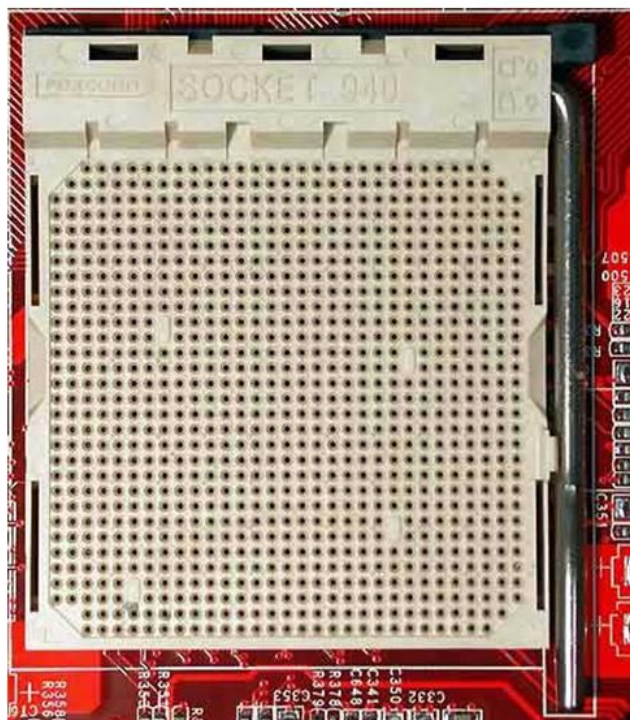
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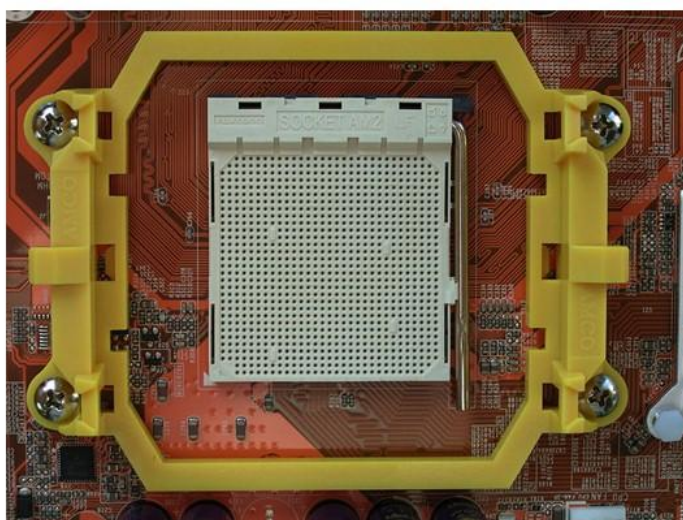
Socket 940

- is a 940-pin socket for 64-bit AMD server processors.
- The more common use of Socket 940 is in the Server market.
- There is a new 940-pin socket called AM2. Despite having the same number of pins, Socket 940 and AM2 are not pin-compatible. This means a Socket 940 processor will not fit in an AM2 socket and vice versa.
- **Processors**
AMD Athlon 64 FX
AMD Opteron



Socket AM2

- are incompatible with 939 motherboards and vice versa, and although it has 940 pins, it is incompatible with [Socket 940](#)
- It was released on May 23, 2006, as a replacement for Socket 939 & Socket 754.
- **Processors** Athlon 64
Athlon 64 X2
Athlon 64 FX
Opteron
Sempron
Phenom



Socket AM2+

- is a CPU socket, which is the immediate successor to Socket AM2 that is currently used in several AMD processors such as Athlon 64 X2. Socket
- is a mid-migration from Socket AM2 to Socket AM3 and is fully compatible with Socket AM2, so processors designed for Socket AM2 will work on Socket AM2+ motherboards and vice versa.
- 940 pins
- Supports Athlon 64, Athlon 64 X2, Athlon 64 FX, Opteron, and Phenom processors.

